

String Indexes

```
In [1]: si = 'Welcome'
        si
```

```
Out[1]: 'Welcome'
```

```
In [2]: si[0]      # Gives the 0th index element
```

```
Out[2]: 'W'
```

```
In [3]: si[4]
```

```
Out[3]: 'o'
```

```
In [4]: si[6]
```

```
Out[4]: 'e'
```

```
In [5]: si[7]
```

```
-----
IndexError                                Traceback (most recent call last)
Cell In[5], line 1
----> 1 si[7]

IndexError: string index out of range
```

```
In [ ]: si[10]
```

```
In [ ]: si[-1]      # Reverse Indexing - Gives the element present in '-1' index
```

```
In [ ]: si[-2]
```

```
In [ ]: si[-7]
```

```
In [ ]: si[-1000]
```

Slicing

(i) Forward Slicing

```
In [6]: si
```

```
Out[6]: 'Welcome'
```

```
In [7]: si[:]      # Gives all elements
```

```
Out[7]: 'Welcome'
```

```
In [8]: si[1:4]      # Gives the element from 1st index to 3rd index (n-1), exclude the 4
```

```
Out[8]: 'elc'
```

```
In [9]: si[0:6]
```

```
Out[9]: 'Welcom'
```

```
In [10]: si[3:7]
```

```
Out[10]: 'come'
```

```
In [11]: si[10]
```

```
-----  
IndexError                                Traceback (most recent call last)  
Cell In[11], line 1  
----> 1 si[10]  
  
IndexError: string index out of range
```

```
In [12]: si[3:10]      # Gives the elements upto last
```

```
Out[12]: 'come'
```

```
In [13]: si[3:100]
```

```
Out[13]: 'come'
```

```
In [14]: si[3:1000]
```

```
Out[14]: 'come'
```

```
In [15]: si[3:5]
```

```
Out[15]: 'co'
```

```
In [16]: si[1:]      # Gives the element from 1st index upto last
```

```
Out[16]: 'elcome'
```

```
In [17]: si[:]
```

```
Out[17]: 'Welcome'
```

```
In [18]: si[3:]
```

```
Out[18]: 'come'
```

```
In [19]: si[:4]      # Gives the element from 0th index from 3rd index (n-1 = 4 - 1 =
```

```
Out[19]: 'Welc'
```

```
In [20]: si[:6]
```

```
Out[20]: 'Welcom'
```

```
In [21]: si = 'Welcome'
```

```
In [22]: si
```

```
Out[22]: 'Welcome'
```

Step Forward Slicing

```
In [23]: si[0:7:2]      # Step Slicing - Jumps 2 elements from 0th index upto 6th index
```

```
Out[23]: 'Wloe'
```

```
In [24]: si[0:7:4]
```

```
Out[24]: 'Wo'
```

(ii) Backward Slicing

```
In [25]: si = 'Welcome'
```

```
In [26]: si
```

```
Out[26]: 'Welcome'
```

```
In [27]: si[-5:-1]
```

```
Out[27]: 'lcom'
```

Step Backward Slicing

```
In [28]: si[-1:-8:-2]
```

```
Out[28]: 'eolW'
```

```
In [29]: si[-2:-7:-2]
```

```
Out[29]: 'mce'
```

```
In [30]: si[-2:-6:-2]
```

```
Out[30]: 'mc'
```

Advance Slicing

```
In [31]: si[::-2]      # Step Slicing - Jumps 2 elements from 0th index upto last
```

```
Out[31]: 'Wloe'
```

```
In [32]: si[::-3]
```

Out[32]: 'wce'

In [33]: `si[::-1]` *# Reverse Slicing*

Out[33]: 'emoclew'

In [34]: `si[::-2]`

Out[34]: 'eolw'

In []: